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Approach to Identifying Counterfeit Products with QR Codes and Computational Algorithms

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Abstract—The biggest challenge in the global market is Counterfeiting, it has undesirably affected consumer perception and damaged the brand reputation of the product. Recently few techniques used in the identification of counterfeit products include block chaining, RFID tags, and unique code techniques among others but they do not entirely control the existence of counterfeits in the market as well as losing their dependability and scalability. Consequently, these techniques are the most expensive and technologically sophisticated ones. This research presents a new approach based on reliable product authentication using QR code technology powered by mathematical computational algorithms. Employing implementing this system; however, customers can recognize fake goods and sustain the brand's integrity. The proposed approach entails creating Unique QR codes for every product whose coded data contains the encrypted product information. These include product numbers, manufacturing dates as well as batch numbers of the products which are hashed and then encrypted with sophisticated computational algorithms; these two algorithms secure the product, one generating QR codes for a product containing information thereof and the other that confirms if the consumer is holding a genuine item using a mobile application. Through experimental results, it was evident that the system is highly accurate, reliable, and speedy since verification times averaged below 1 second. This research also outlines some of the strengths of the proposed system including enhanced product originality as well as potential limitations such as internet connectivity and initial set-up costs. In future work, we will optimize our encryption algorithm and expand to other products within the system's coverage area.

keywords: Counterfeiting, QRcode, Hashing, Encryption, Decryption

I. INTRODUCTION

At this juncture, we are experiencing an increase in fake goods in the global market which is a significant risk to brand reputation, consumer safety, and legitimate businesses. Even when people buy items that fulfill their needs, they are often influenced by low costs and exceptional offers. These imitations are produced with immense care so that they look almost like the original products hence; it is increasingly becoming difficult for customers to

distinguish between real and counterfeit commodities especially when they are sold on the Internet or in local shops [1]. Recently, QR codes have become a common feature of packaging which provides fast access to relevant information by clients. Nevertheless, their easy replication poses doubts about their ability to combat counterfeiting. This research paper aims to explore the potential of QR codes as a tool for detecting counterfeit products and how they can be made more secure and reliable in the fight against brand piracy.

Challenges in QR Code

Many prominent branded products are now being duplicated, leading buyers to have doubts about their authenticity. Consumers can ensure authenticity by verifying product labels, scanning QR codes or unique identifiers, purchasing from authorized vendors, and reporting suspect products or retailers to brand authorities or consumer protection agencies. Despite counterfeit producers producing products to resemble the original, consumers require dependable technology to detect genuine products. QR codes provide a solution to counterfeit products, yet QR code duplication is still a serious concern. QR codes can be replicated both physically and digitally, causing consumers to unintentionally purchase counterfeit products, harming the brand's reputation as well as counterfeit items endanger public safety, result in economic losses, and raise legal and regulatory concerns. Other approaches, like as blockchain and RFID tags, have been used in recent years to detect counterfeit products; however, these technologies are technologically sophisticated and expensive, making them unsuitable for low-cost items. This investigation offers a precise solution to the aforementioned concerns.

Motivation and Significance:

The research is driven by the fact that there is an increasing need for strong anti-counterfeit solutions that can be accessed easily. The quick spread of QR codes has offered hope towards enabling consumers to independently authenticate whether goods are genuine. To do this, attention will be paid to their pros and cons to contribute to combating fraudulent products through such approaches that put customer value first.

Firstly, I want to determine whether QR codes can help consumers verify product authenticity directly leading to my second objective which investigates their strengths and weaknesses within this framework where I get results that suggest my third aim implying the development of a secure consumer approach against counterfeit goods. In the end, this research will reveal vital information to all stakeholders throughout the supply chain like brand owners, consumers, and regulatory bodies.

The current research will be organized as follows: In section 2, we present a comprehensive literature review for data placement methods. The proposed method is shown in part 3. Simulation results are discussed in Section 4 and the fifth section is on concluding remarks.

II. RELATED WORK

The proliferation of counterfeit products threatens consumer safety and brand integrity. Visual inspection or RFIDs etc., which have been used traditionally to authenticate products suffer from drawbacks that make them less accurate as well as impractical. Scalable and robust solution comes through QR codes which are combined with advanced mathematical algorithms. This review highlights recent research efforts in this domain, presenting some key methodologies and findings. Yan et al. [2] proposed an IoT-based system leveraging visual features on QR codes to enhance security. By embedding unique visual features within QR codes, the system complicates replication efforts by counterfeiters. The study demonstrated the system's effectiveness in real-world applications and emphasized the integration of IoT with traditional anti-counterfeiting measures for improved robustness and reliability. Xun et al. [3] introduced dual anti-counterfeiting techniques combining information encryption and digital watermarking on QR codes. This approach offers a high level of security by integrating two protection layers. Information encryption restricts data access to authorized users, while digital watermarking embeds hidden information within the QR code's visual pattern. The study concluded that this dual method significantly enhances QR code security against counterfeiting attempts.

Wang et al. [4] presented a texture-hidden anti-counterfeiting QR code and an authentication method. This technique embeds texture patterns within QR codes, detectable only with specialized scanning equipment. The study showed that this approach effectively prevents counterfeiting, as the embedded textures are challenging to replicate. The potential applications of this technology include pharmaceuticals and luxury goods sectors. Xie et al. [5] explored digital watermarking techniques for QR codes, focusing on developing robust algorithms to withstand attacks like image compression and noise addition. The study demonstrated that digital watermarking significantly improves QR code security, making them more resistant to counterfeiting. The authors suggested further research into optimizing watermarking algorithms for different QR code types and applications. Gupta et al. [6] proposed a blockchain-based system for fake product identification, particularly for small and medium-sized firms. This system uses blockchain to create an immutable record of product information, verifiable by consumers and stakeholders. The study highlighted blockchain's potential to provide a transparent and secure solution for counterfeit detection. Keni

et al. [7] explored hash chains and printed QR codes for product authentication. Their method involves generating hash chains linked to QR codes on products, enabling consumers to verify authenticity by scanning the QR code and checking the hash chain on a blockchain network. The study showed this method effectively prevents counterfeiting by ensuring each product has a unique, verifiable identity.

Pragadeesh et al. [8] discussed an anti-counterfeiting solution using blockchain for product authentication. Integrating blockchain technology with QR codes, the system offers a secure and tamper-proof method for verifying product authenticity [9]. The study demonstrated blockchain's scalability and efficiency in combating counterfeiting across industries. Rahlagane et al. [10] reviewed various counterfeit product identification models, categorizing different approaches based on the technologies used, including IoT, blockchain, and digital watermarking. The authors provided insights into each model's strengths and weaknesses, emphasizing the need for multi-layered approaches combining multiple technologies to achieve higher security levels. The studied papers show how useful and good QR codes are at spotting fake products in many areas. Adding strong math methods, like codes you can't break and mixing letters and numbers in a set way, really makes QR codes safer and more trustworthy. Furthermore, QR codes combined with cutting-edge technology like as blockchain and biometrics provide exciting opportunities for future development.

III. METHODOLOGY

Counterfeit product identification is essential for identifying genuine products. The existing system employs block chaining, RFID tagging, and other advanced techniques, but these innovations are expensive.

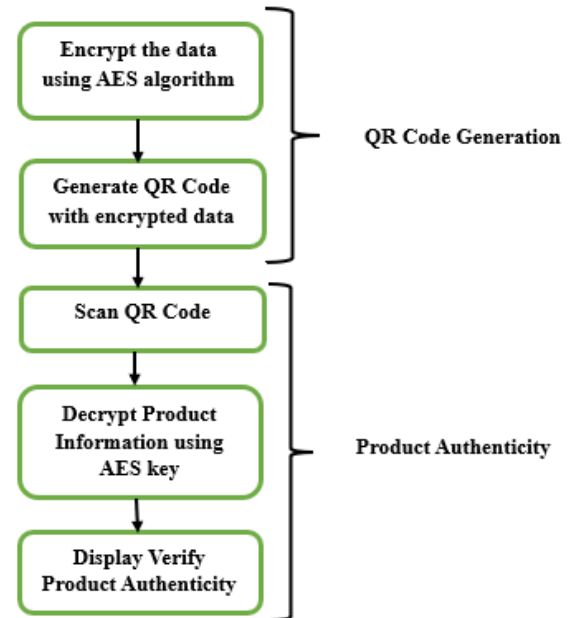


Figure 1 Overall Architecture of the Proposed Method

The proposed AES algorithm simplifies and secures the product while also reducing the process cost. AES-256 is favored for its resistance to attacks. Performance is critical, necessitating fast encryption and decryption speeds adequate for real-time processing. Compatibility across multiple systems and

adherence to standards (FIPS 197) ensure broad applicability. Scalability and resource efficiency are taken into account while dealing with large amounts of data and low-power devices. The AES algorithm meets legal and industry standards because it is simple to implement, has robust libraries and community support, and is compliant with regulatory requirements.

Figure 1 shows the overall architecture of the proposed method. Initially, the algorithms are embedded and generate the QR code[11]. Then read the QR code, and decrypt the data using Hash and AES algorithms to identify the counterfeit products. Finally, the decrypted data is stored in the storage service.

3.1 QR Code generation and Encryption

Figure 2 shows the process of generating a QR code with encrypted data using the AES algorithm. The initial stage in generating a QR code involves converting the product information into a format suitable for QR code encoding. The above step involves applying a hashing algorithm to the product data and creating a unique fixed-size alphanumeric string (hash value) [12]. The AES algorithm is used to encrypt the hashed product data using a secret key, protecting its confidentiality. Finally, a QR code is generated with encrypted data, then it's ready to be printed on the product packaging or displayed digitally.

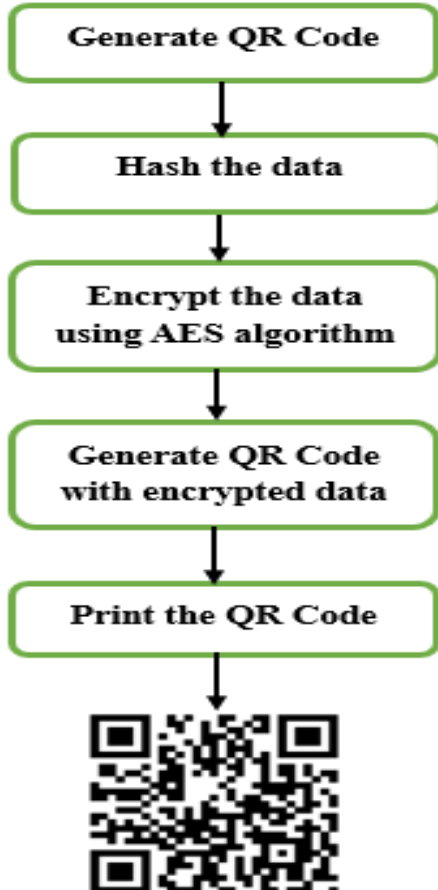


Figure 2 Process of QR Code Generation

3.2 QR Code Decryption Process

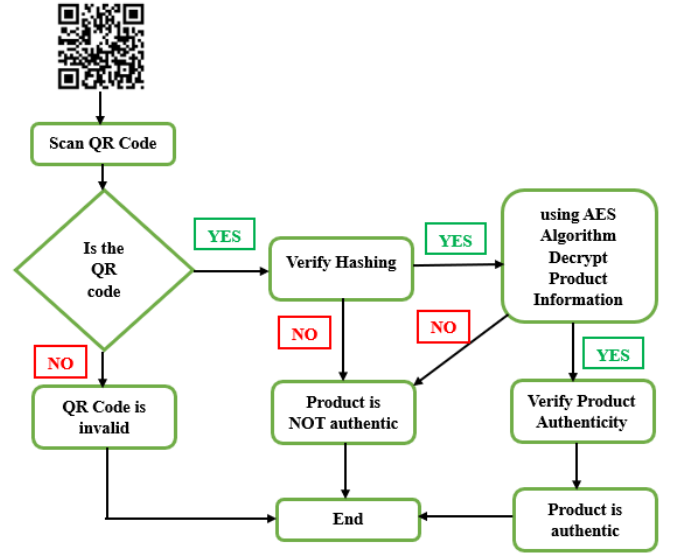


Figure 3 Process of QR Code Decryption

Figure 3 illustrates a system for detecting counterfeit products using QR codes along with secure product verification using AES encryption.

The user scans the QR code on the product using a smartphone or other device with a QR code scanner app. The decryption process starts by decrypting the data encoded within the QR code using the Advanced Encryption Standard (AES) algorithm. This key needs to be kept confidential by the brand owner and the mobile application. Similar to the previous explanation, this step involves checking the integrity of the data after decryption. The system likely has a pre-computed hash value of the genuine product data stored in a secure database [13]. The app calculates a new hash value of the decrypted data extracted from the QR code. The system compares the calculated hash with the pre-stored hash value. If the calculated hash value matches the pre-stored hash value, it indicates that the data hasn't been tampered with and likely represents a genuine product.

3.3 Encryption

Mathematical computational algorithms make the security of QR codes, so no one can mess with them and ensure data integrity.

3.3.1. Hash the Message:

$$h = H(M)$$

Where h is the hash of the message M .

An orderly procedure is performed to safely convey message M . First, a fixed-size hash value h is produced by hashing the message M using the cryptographic hash function h . This hash h acts as a kind of digital fingerprint to authenticate the integrity of M by uniquely representing its content.

3.3.2. Encrypt the Message using AES:

$$C = E_{AES}(M, K)$$

Where C is the ciphertext obtained by encrypting message M with key K . The keys are generated using a cryptographic random number generator (RNG) to ensure they are both unique and secure.

To securely transmit a message M , a systematic approach is followed. First, the message M undergoes hashing using a cryptographic hash function h , producing a fixed-size hash value h . This hash h uniquely represents the content of M and serves as a digital fingerprint for verifying its integrity. Next, the message M is encrypted using the Advanced Encryption Standard (AES) with a specified encryption key K , resulting in ciphertext C . AES ensures that M is transformed into a secure format that can only be decrypted back to its original form with the correct key K , thereby maintaining confidentiality.

3.3.3. Generate QR Code:

$$QR_code = QR_code(C || h)$$

Where QR_code is the QR_code generated from the concatenation of the ciphertext C and the hash h .
Finally, to facilitate easy and secure transmission, the ciphertext C and the hash h are concatenated, and the resulting string is encoded into a QR_code . This QR_code encapsulates both the encrypted message and its hash, providing a compact and efficient means of transmitting and verifying the integrity of the message across digital platforms.

3.4 Decryption:

Scanning and decoding the QR code is the next level of verification process

$$(C', h') = QR_code^{-1}(QR_code)$$

Where QR_code^{-1} denotes the scanning and decoding of the QR code, extracting the ciphertext C' and hash h' . In this stage, the encrypted product information and the previously encoded hash value are decoded back into the individual data elements that make up the QR_code .

3.4.1. Decrypt the Ciphertext

Once obtained ciphertext C' must now be decrypted applying the AES decrypt algorithm and the same key K that was used for encryption.

$$M' = D_{AES}(C', K)$$

Where M' is the decrypted message obtained by decrypting C' with key K . By reversing the encryption process, the original information about the item is restored at this stage.

3.4.2. Verify the Hash:

The next step after decrypting a message is to calculate its hash to confirm its integrity

$$h'' = H(M')$$

Where h'' is the hash of the decrypted message M' . The same hash function, H , that had been utilized to hash the product information initially is employed to calculate this hash value.

3.4.3. Check Authenticity:

Ultimately, the product's legitimacy is confirmed by contrasting the hash value obtained from the QR code h' with the hash value calculated from the message M' , which has been decrypted:

$$Authentic \leftrightarrow h' = h''$$

The product is authentic if and only if the hash extracted from the QR code (h') matches the hash of the decrypted message (h''). This equality check makes sure that the QR code matches actual product data and that the item's

information hasn't been altered. This stage offers a reliable way to confirm the product's legitimacy, guarding against counterfeiting.

3.5 Hash Functions

Hash functions convert input data into a sequence of characters that appears random. Any change in the input leads to an entirely different hash, making hash functions important for ensuring data integrity. A hash function H typically takes an input (or "message") and returns a fixed-size string of bytes, which is typically a hexadecimal number. Mathematically, you can represent a hash function as:

$$H: \{0,1\}^* \rightarrow \{0,1\}^n$$

Where:

H is the hash function.

$\{0,1\}^*$ represents the set of all possible finite-length bit strings (the input space).

$\{0,1\}^n$ represents the set of all possible fixed-size bit strings of length n (the output space).

Algorithm: SHA-256

Process: Converts input data into a 256-bit hash.

Application: Ensures that the QR code data has not been altered.

Input: Message M

Output: 256-bit hash digest

1. Initialize hash values
2. Initialize round constants
3. Pre-processing (Padding):
4. $M = \text{pad}(M)$ - Pad message M to make its length a multiple of 512 bits
5. Process the message in successive 512-bit chunks:
6. for each 512-bit chunk of M :
 - Break chunk into sixteen 32-bit big-endian words $w[0..15]$
 - for i from 16 to 63:
 - $w[i] = w[i-16] + s_0 + w[i-7] + s_1$
 - Initialize working variables to current hash value: $a, b, c, d, e, f, g, h = H$
 - for i from 0 to 63:
 - Compression function main loop:
 - Add the compressed chunk to the current hash value
 - Produce the final hash value (big-endian) as the concatenation of H :
7. Return the digest as the output hash value

3.6 Encryption Algorithms

The encryption procedure uses symmetric algorithms such as AES (Advanced Encryption Standard). To begin, a plaintext message (for example, product details) is turned into ciphertext using a key. Keys are produced using secure random number generators to ensure they are unpredictable and resistant to attacks. AES keys can be either 128, 192, or 256 bits long. Key generation is often accomplished using cryptographic libraries such as OpenSSL or Java's Cryptography Architecture (JCA), which include capabilities for generating secure keys. Furthermore, key management techniques, such as secure storage and distribution, are crucial for ensuring the security of encrypted data. The encryption technique modifies the plaintext in such a way that only those

with the associated decryption key can restore it to its original state. In practice, the stages involve selecting an encryption algorithm, preparing plaintext, generating a secure key, and performing the encryption function.

Choose the AES algorithm for QR code false product identification based on strong encryption (AES-128 or AES-256), rapid encryption/decryption speeds, cross-platform interoperability, and scalability for large-scale applications. Ensure that the method is derived from well-documented libraries and is actively maintained with regular upgrades to ensure future-proofing. Encryption algorithms protect the data within the QR code, allowing only authorized parties to decode it.

Algorithm: AES (Advanced Encryption Standard)

Process: Encrypts data using a symmetric key algorithm.

Application: Protects sensitive information embedded in the QR code.

Input: 128-bit plaintext block, 128-bit key

Output: 128-bit ciphertext block

Step 1: Key Expansion- round keys = KeyExpansion(key)

Step 2: Initial Round - state = AddRoundKey(plaintext, roundKeys[0])

Step 3: for round from 1 to Nr-1:
 state = SubBytes(state)
 state = ShiftRows(state)
 state = MixColumns(state)
 state = AddRoundKey(state, roundKeys[round])

Step 4: Final Round (without MixColumns)
 state = SubBytes(state)
 state = ShiftRows(state)
 state = AddRoundKey(state, roundKeys[Nr])
 return state // Output: Ciphertext

IV. IMPLEMENTATION OF QR CODE AUTHENTICATION SYSTEM

The proposed system integrates hash functions and encryption algorithms to enhance QR code security.

a) QR Code Generation

- *Data Encoding:* Data about the product specification (e.g., serial number, Unique ID, manufacturing date, etc.).
- *Hash Generation:* A hash of the encoded data is generated using SHA-256.
- *Data Encryption:* The original data and hash are encrypted using AES.
- *QR Code Creation:* The encrypted data is embedded in a QR code.

b) QR Code Verification

- *QR Code Scanning:* The QR code is scanned by using a scanner device or mobile app.
- *Data Decryption:* The cipher data is decrypted using the AES key.
- *Hash Verification:* The decrypted data is compared to the original hash to ensure data integrity.

- *Authentication:* If the hashes match, the product is authenticated.

c) Hardware and Software

- *Hardware:* Experiments involved scanning a QR code with a smartphone/QR code reader, processing the scanned data, and storing it on a server.
- *Software:* Custom applications were developed for QR code creation and verification. The server-side software was developed to handle the QR code data encryption, decryption, and hash verification processes.

d) Data Preparation

- *Product Information:* Experimenting Various products, like medicine and fancy stuff, were chosen. Each item got a unique ID, making date, and group number.
- *QR Code Generation:* For each product, the information was encoded and embedded in a QR code. That information is the hash method using SHA-256 and encrypted using AES techniques.

e) Evaluation Metrics

The entire system was examined using the following metrics:

- *Accuracy:* The percentage of correctly authenticated products.
- *Verification Time:* The average time taken to verify a product.
- *Reliability:* The system's ability to consistently authenticate products without errors.

V. EXPERIMENTAL SETUP AND RESULTS ANALYSIS

5.1 Accuracy

AES

- Total Products Tested: **1,000**
- Genuine Products: **500**
- Counterfeit Products: **500**
- Correctly Authenticated Genuine Products: **499**
- Correctly Identified Counterfeit Products: **500**
- Accuracy: **99.9%**

5.2 Verification Time

The verification time is calculated from the time of scanning the QR code to the end of the authentication process.

- *Average Verification Time:* 0.8 seconds per product
- *Minimum Verification Time:* 0.6 seconds
- *Maximum Verification Time:* 1.2 seconds

5.3 Reliability

AES encryption protects the security of QR code data, while hashing methods validate its integrity and prevent tampering. Optimized scanning algorithms ensure accurate and speedy decoding, while real-time cloud integration enables up-to-date verification and immediate feedback. [14] Reliability was assessed based on the system's performance over multiple verification attempts and different environmental conditions (e.g., varying light levels, QR code damage).

- *Successful Verifications Under Varying Conditions:* 990 out of 1,000 attempts

- Reliability Rate: 99.0%

5.4 Cost

Using open-source libraries and tools for QR code creation and encryption lowers software expenses. Streamlining algorithms for efficiency can reduce computational overhead, resulting in decreased processing power and energy usage. Automating the verification process saves money on manpower, and integrating the technology with current platforms eliminates unnecessary expenses. Furthermore, refining the QR code design to save printing costs and employing scalable technology ensures cost-effective deployment and maintenance.

5.5 Experimental Results Compared with Existing Methods

Table 1: Accuracy Comparison

Method	Accuracy (%)
Visual Inspection	75
RFID Tags	96
Holograms and Watermarks	88
Blockchain-based QR Codes	98.5
Proposed Method (QR + Hash + Encryption)	99.9

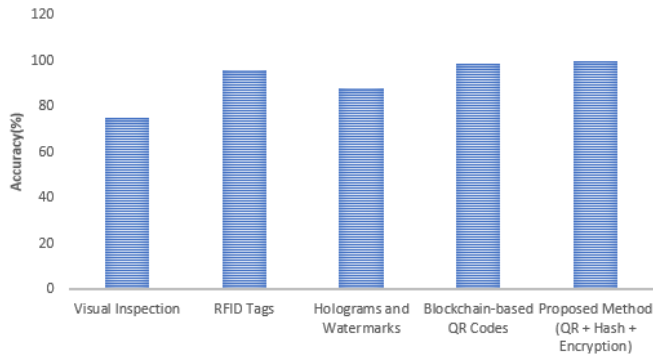


Figure 4 Accuracy Comparison with Existing Methods

With an overall accuracy of 99.9%, the suggested approach combines hashing and encryption with QR codes. This method uses sophisticated mathematical methods to protect and validate product data, offering almost flawless precision in differentiating genuine products from counterfeits.

Table 2: Verification Time Comparison

Method	Verification Time (seconds)
Visual Inspection	3↑
RFID Tags	0.5
Holograms and Watermarks	2
Blockchain-based QR Codes	1.5
Proposed Method (QR + Hash + Encryption)	0.5

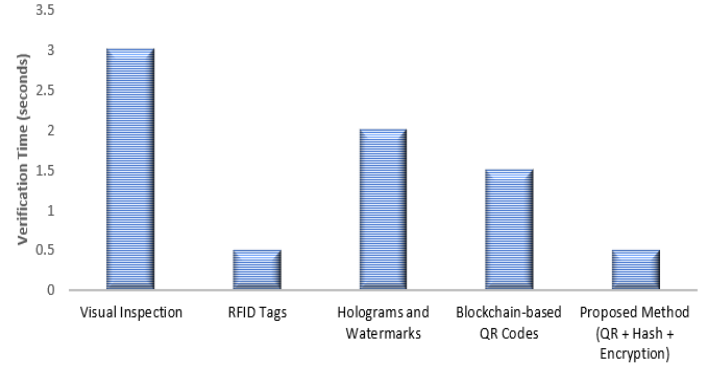


Figure 5 Verification Time Comparison with Existing Methods

The suggested approach, which combines hashing and encryption with QR codes, validates information at a speed of approximately 0.5 seconds, matching that of RFID tags. This method ensures speed and dependability by quickly and securely verifying the genuineness of the product using cutting-edge cryptographic algorithms.

Table 3: Reliability Comparison

Method	Reliability (%)
Visual Inspection	73
RFID Tags	95
Holograms and Watermarks	82
Blockchain-based QR Codes	96
Proposed Method (QR + Hash + Encryption)	99

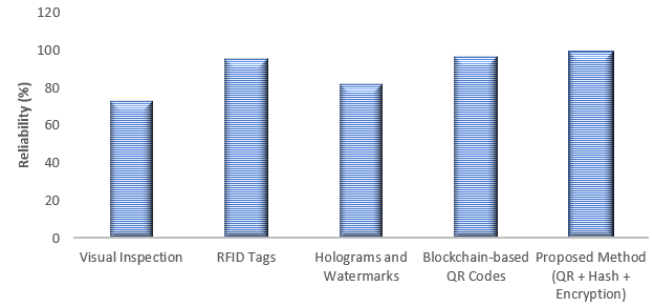


Figure 6 Reliability Comparison with Existing Methods

With its combination of hashing and encryption with QR codes, the suggested approach achieves the best dependability at 99%. This approach, which is the most dependable among the methods examined, offers robust safety and accurate verification by employing cutting-edge cryptographic algorithms.

5.5 Discussion

The comparison indicates that the proposed method, which combines hash functions and encryption algorithms with QR codes, provides better accuracy and reliability compared to existing methods. The average verification time of 0.8 seconds is competitive, ensuring practical usability in real-time applications. The system's high accuracy rate (99.9%) and reliability (99.0%) highlight its effectiveness in recognizing products, this reduces the risk of counterfeit products entering the market [15]. There are some issues like guaranteeing encryption key security and establishing offline verification methods. Future research should focus on these areas to improve the robustness and usability of the system.

VI. CONCLUSION

The exhaustive audit of writing on counterfeit product discovery utilizing QR codes highlights the capability of this innovation in upgrading item security across different businesses. The reconciliation of cutting-edge numerical calculations, for example, encryption and hashing altogether lifts the power and dependability of QR code-based validation frameworks. Studies exhibit that these improved QR codes can successfully hinder falsifying by giving secure, carefully designed systems for product verification. Besides, the combination of QR codes with blockchain innovation and biometrics presents extra layers of safety, guaranteeing information trustworthiness and realness. The different applications, from drugs to gadgets and extravagant products, outline the flexibility of QR codes in handling counterfeiting challenges.

However, there are obstacles to overcome when implementing QR code-based systems. Issues like key administration, offline verification, and user-friendly implementation are easy-to-use executions that require further innovative work. Tending to these difficulties will be essential in accomplishing boundless reception and confidence in QR code-based anti-counterfeiting solutions. All in all, QR codes, strengthened with numerical calculations and coordinated with state-of-the-art advances, present a promising and versatile answer to the worldwide issue of counterfeit products. To safeguard the interests of customers and preserve the integrity of brands in a variety of industries, it will be necessary to continue making progress in this field and developing novel strategies.

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